

Week 8: Learning – Maximum Likelihood Estimation

Overview, Resources and Notes

1 Overview

So far in this module, we have looked at *modelling* probabilistic systems and performing *inference* using these models. When constructing models, we have so far constructed models by hand, and we have assumed that all the information necessary to construct a model is available. However, defining a large, complex model completely by hand is tedious at best and practically impossible at worst. For this reason, we now move on to *learning* models (or certain aspects of models) from data.

Learning in PGMs is usually separated into the following two tasks:

- *Structure learning* aims to determine the RVs and the way they are connected together in the graph.
- *Parameter learning* aims to determine the parameters of the factors required by the graph (such as the entries in a discrete table, or the covariance matrix and mean vector of a Gaussian factor).

In this module, we only focus on parameter learning. We therefore assume that the RVs and the graph connecting the RVs (either a Bayes network (BN) or Markov network (MN)) are given, and we aim to learn the parameters describing the factors in the model.

There are different variations of the parameter learning problem. We focus on the following:

- Although parameter learning in BNs and MNs broadly follow the same approach, the partition function in MNs creates issues that requires more advanced techniques to handle. For this reason, we only look at parameter learning for BNs in this module.
- For this week and next week, we assume that the data consists of observations for *all* the RVs in the model. In Week 10, we will relax this assumption and look at the more challenging case where some RVs in the model are latent (unobserved).
- For this week, we aim to find *values* for the parameters (i.e., we aim to find point estimates). We also aim to find the parameter values that maximise the likelihood of the data (this is the so-called *maximum likelihood estimation (MLE)*). We will look at alternative approaches to both of these aspects next week.

Given the context of learning in PGMs and the approach we take in this week, let us now look in a bit more detail what maximum likelihood estimation (MLE) for BNs with only observed RVs involves. One data point is a collection of observed values for all the RVs in our model, and it is denoted by $\mathbf{d}^{[m]}$. We assume that there are M data points; i.e., the data is given by $\mathcal{D} = (\mathbf{d}^{[1]}, \dots, \mathbf{d}^{[M]})$. All the parameters in the model are denoted by Θ .

MLE aims to find values for the model parameters Θ in such a way that the total likelihood of the data $\mathcal{D}$ given the parameters is maximised; i.e., it finds

$$\Theta^{\text{ML}} = \arg \max_{\Theta} p(\mathcal{D}|\Theta). \quad (1)$$

This is usually done by finding where the partial derivative of the likelihood function with respect to the parameters is equal to zero. One can obtain the same result by calculating

$$\Theta^{\text{ML}} = \arg \max_{\Theta} \log p(\mathcal{D}|\Theta), \quad (2)$$

since $\log(\cdot)$ is a strictly monotonic function – this formulation is often more convenient to use. Note that Θ (and not $\mathcal{D}$) is the argument to p , changing it from a probability distribution to a *likelihood* function. In spite of its simplicity, this technique is very useful and often forms a building block of more involved techniques.

We also assume that the data points are independent and identically distributed (i.i.d.), which means that we can factorise the data likelihood as

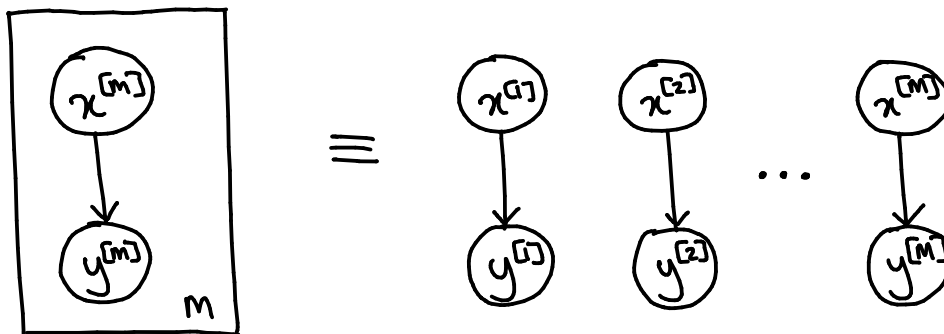
$$p(\mathcal{D}|\Theta) = \prod_{m=1}^M p(\mathbf{d}^{[m]}|\Theta), \quad (3)$$

which means that we can decompose the log-likelihood function as

$$\log p(\mathcal{D}|\Theta) = \log \prod_{m=1}^M p(\mathbf{d}^{[m]}|\Theta) = \sum_{m=1}^M \log p(\mathbf{d}^{[m]}|\Theta). \quad (4)$$

This is as far as we can go for the general case; for every specific problem, we have to do the rest of the calculations from scratch. See Sections 3 and 4 for two worked-out examples – one for discrete RVs and one for Gaussian RVs.

To represent our model across all data points, we use plate notation, which is a concise way to represent models (or parts of models) that are repeated. For example, a simple BN with two RVs, $x^{[m]}$ and $y^{[m]}$, for M data points is shown below on the left-hand side, with the equivalent set of models drawn without plate notation shown on the right-hand side.



For this example model, one data point is given by $\mathbf{d}^{[m]} = (x^{[m]}, y^{[m]})$.

2 Resources

Prescribed resources

Study the following resources before the lectures and practical on Thursday (see the notes further down to get an idea of what to focus on):

- **Barber** Chapter 8, Section 9.3, and the introduction to Section 9.6
- **Murphy** Section 4.2.8
- **Bishop** Section 8.1.1
- The **PGM Tutorial** Sections 6.1 and 6.2.1

Supplementary resources (optional)

I also recommend that you study the following resources, if you have access to it:

- **Koller videos** in the Coursera course Probabilistic Graphical Models 1: Representation, Week 2:
 - *Template Models: Plate Models (20:08)*
- **Koller videos** in the Coursera course Probabilistic Graphical Models 3: Learning, Week 1:
 - *Learning: Overview: Learning: Overview (15:35)*

- *ML-class Revision (Optional)* (by Andrew Ng) from *Regularization: The Problem of Overfitting* (9:42) to *Regularization and Bias Variance* (11:20)
- *Maximum Likelihood Parameter Estimation in BNs: Maximum Likelihood Estimation* (14:59)
- *Maximum Likelihood Parameter Estimation in BNs: Maximum Likelihood Estimation for Bayesian Networks* (15:49)
- **Koller videos** in the Coursera course Probabilistic Graphical Models 3: Learning, Week 2:
 - *Parameter Estimation in MNs: Maximum Likelihood for Log-Linear Models* (28:47)
 - *Parameter Estimation in MNs: Maximum Likelihood for Conditional Random Fields* (13:24)

If you would like to read further, the following resources might be helpful: **Koller&Friedman** Chapter 16 and Sections 17.1 and 17.2; **Kochenderfer** Section 4.1.

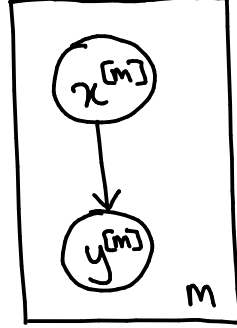
Comments on the resources

- **Barber** Chapter 8 establishes the foundations of parameter estimation, and Section 9.3 applies this foundation to MLE for BNs. Specifically, take note of the following:
 - Section 8.1: Take note of the different forms of data encoding, especially the 1-of- m encoding in Subsection 8.1.1.
 - Section 8.2: You should already be familiar with the concepts in the first half of this section. Take note of the following concepts; you can just skim over the rest:
 - * The empirical distribution describes the observed data and has use in parameter estimation. Note the use of Kronecker and Dirac delta functions to cover discrete and continuous spaces respectively.
 - * The Kullback-Leibler divergence measures the “difference” between two distributions. It plays an important role in some objective functions.
 - Sections 8.3-8.5: These sections contain a mixture of familiar concepts and concepts that we do not focus on in this week; you can just skim through these sections.
 - Section 8.6: Although this section contains both ML and Bayesian parameter estimation techniques (we’ll get to the latter in Week 9), the perspective of learning given by this section is important.
 - Section 8.7 discusses the properties of ML estimation and is important to understand. ML estimation is equivalent to minimising the Kullback-Leibler divergence between the empirical distribution and the distribution that we want to fit to the data. We will later see that fitting one distribution to another via the Kullback-Leibler distance, also has other uses in estimation.
 - Section 8.8 discusses learning the parameters of a Gaussian distribution. ML estimation of a Gaussian distribution (Subsection 8.8.1) is the important part to understand (for now)¹; you can ignore the rest.
 - Section 9.3: This section about ML parameter estimation for BNs is important – it sets out the approach to parameter estimation we do in this week.
- The introduction to **Barber** Section 9.6 (up to the start of Section 9.6.1) gives an overview of MLE in MNs, and gives the reason why MLE is typically more challenging in MNs than in BNs. Although we do not cover parameter learning in MNs in this module, it is important that you understand the reason why it is difficult.
- **Murphy** Section 4.2.8 and **Bishop** Section 8.1.1 (specifically Figures 8.4 to 8.7) explain plate notation, which is used to depict graphs with repeated structures. Plate notation is regularly used in the material and practicals of Weeks 8 to 10.
- The **PGM Tutorial** Sections 6.1 and 6.2.1 cover some of the same ground as the material from **Barber**, but also some new ground. Note the following specific aspects:
 - Similar to **Barber**, Section 6.1 also interprets MLE as the maximisation of the empirical distribution, but its perspective of the learning task and the discussion of the bias-variance trade-off are different than in **Barber**, and are important to understand.
 - Section 6.2.1 gives a nice concise overview of MLE for BNs (for discrete RVs) that mostly overlaps with **Barber**.

¹Equation 8.8.3 in the 2025 version of Barber’s textbook contains an error: the RHS should be $N\Sigma^{-1}\mu$.

3 Example with discrete RVs

Let us now work through two simple examples that use MLE in BNs. The first example uses the following model drawn in plate notation:



We assume that the two random variables are binary; i.e., $x^{[m]} \in \{0, 1\}$ and $y^{[m]} \in \{0, 1\}$. The two factors for this model are given by the following tables:

$x^{[m]}$	$p(x^{[m]})$	$x^{[m]}$	$y^{[m]}$	$p(y^{[m]} x^{[m]}, \theta_e)$
0	0.5	0	0	$1 - \theta_e$
1	0.5	0	1	θ_e
		1	0	θ_e
		1	1	$1 - \theta_e$

This means that we can think of $y^{[m]}$ as an error-prone measurement of $x^{[m]}$, where the probability of an incorrect measurement is given by the parameter $\theta_e \in [0, 1]$. There is only one parameter; i.e., $\Theta = (\theta_e)$.

To calculate the maximum-likelihood estimate for θ_e , it is convenient to rewrite the factors of the model in terms of Iverson brackets. One can think of Iverson brackets as an indicator function for a logical statement, and it is the same as the Kronecker delta function. For example, the Iverson brackets applied to the logical statement P is defined as

$$\llbracket P \rrbracket = \begin{cases} 1, & P \text{ is true} \\ 0, & P \text{ is false.} \end{cases} \quad (5)$$

Using Iverson brackets, we can now rewrite the first factor as

$$p(x^{[m]}) = 0.5 \llbracket x^{[m]}=0 \rrbracket \cdot 0.5 \llbracket x^{[m]}=1 \rrbracket, \quad (6)$$

and the second factor as

$$\begin{aligned} p(y^{[m]}|x^{[m]}, \theta_e) &= (1 - \theta_e) \llbracket x^{[m]}=0, y^{[m]}=0 \rrbracket \cdot \theta_e \llbracket x^{[m]}=0, y^{[m]}=1 \rrbracket \\ &\quad \cdot \theta_e \llbracket x^{[m]}=1, y^{[m]}=0 \rrbracket \cdot (1 - \theta_e) \llbracket x^{[m]}=1, y^{[m]}=1 \rrbracket \\ &= (1 - \theta_e) \llbracket x^{[m]}=0, y^{[m]}=0 \rrbracket + \llbracket x^{[m]}=1, y^{[m]}=1 \rrbracket \cdot \theta_e \llbracket x^{[m]}=0, y^{[m]}=1 \rrbracket + \llbracket x^{[m]}=1, y^{[m]}=0 \rrbracket. \end{aligned} \quad (7)$$

In the last line, the exponent $\llbracket x^{[m]}=0, y^{[m]}=0 \rrbracket + \llbracket x^{[m]}=1, y^{[m]}=1 \rrbracket$ will only be 1 if $x^{[m]} = y^{[m]}$, and 0 otherwise. Similarly, the exponent $\llbracket x^{[m]}=0, y^{[m]}=1 \rrbracket + \llbracket x^{[m]}=1, y^{[m]}=0 \rrbracket$ will only be 1 if $x^{[m]} \neq y^{[m]}$, and 0 otherwise. We can therefore rewrite the second factor as

$$p(y^{[m]}|x^{[m]}, \theta_e) = (1 - \theta_e) \llbracket x^{[m]}=y^{[m]} \rrbracket \cdot \theta_e \llbracket x^{[m]} \neq y^{[m]} \rrbracket. \quad (8)$$

Let's first write the log-likelihood function of the data in terms of the factors of our model:

$$\begin{aligned}
 \log p(\mathcal{D}|\Theta) &= \log \prod_{m=1}^M p(\mathbf{d}^{[m]}|\Theta) \\
 &= \sum_{m=1}^M \log p(\mathbf{d}^{[m]}|\Theta) \\
 &= \sum_{m=1}^M \log \left[p(y^{[m]}|x^{[m]}, \theta_e) p(x^{[m]}) \right] \\
 &= \sum_{m=1}^M \left[\log p(y^{[m]}|x^{[m]}, \theta_e) + \log p(x^{[m]}) \right].
 \end{aligned} \tag{9}$$

Now we substitute in the factors rewritten in terms of the Iverson brackets:

$$\begin{aligned}
 \log p(\mathcal{D}|\Theta) &= \sum_{m=1}^M \left\{ \log \left[(1 - \theta_e)^{\mathbb{I}[x^{[m]}=y^{[m]}]} \cdot \theta_e^{\mathbb{I}[x^{[m]} \neq y^{[m]}]} \right] + \log \left[0.5^{\mathbb{I}[x^{[m]}=0]} \cdot 0.5^{\mathbb{I}[x^{[m]}=1]} \right] \right\} \\
 &= \sum_{m=1}^M \left\{ \mathbb{I}[x^{[m]} = y^{[m]}] \cdot \log(1 - \theta_e) + \mathbb{I}[x^{[m]} \neq y^{[m]}] \cdot \log \theta_e \right. \\
 &\quad \left. + \mathbb{I}[x^{[m]} = 0] \cdot \log 0.5 + \mathbb{I}[x^{[m]} = 1] \cdot \log 0.5 \right\}.
 \end{aligned} \tag{10}$$

At this point, we are ready to find the value of the parameter θ_e that maximises the log-likelihood function of the data. To do this, we take the partial derivative of the log-likelihood function with respect to the parameter θ_e , and we set the result to 0:

$$\begin{aligned}
 &\frac{\partial}{\partial \theta_e} \log p(\mathcal{D}|\Theta) \Big|_{\theta_e = \theta_e^{\text{ML}}} = 0 \\
 \Rightarrow &\sum_{m=1}^M \left\{ \mathbb{I}[x^{[m]} = y^{[m]}] \cdot \frac{-1}{(1 - \theta_e^{\text{ML}})} + \mathbb{I}[x^{[m]} \neq y^{[m]}] \cdot \frac{1}{\theta_e^{\text{ML}}} + 0 \right\} = 0 \\
 \Rightarrow &\sum_{m=1}^M \left\{ -\mathbb{I}[x^{[m]} = y^{[m]}] \cdot \theta_e^{\text{ML}} + \mathbb{I}[x^{[m]} \neq y^{[m]}] \cdot (1 - \theta_e^{\text{ML}}) \right\} = 0 \\
 \Rightarrow &\sum_{m=1}^M \mathbb{I}[x^{[m]} \neq y^{[m]}] = \theta_e^{\text{ML}} \sum_{m=1}^M \left\{ \mathbb{I}[x^{[m]} = y^{[m]}] + \mathbb{I}[x^{[m]} \neq y^{[m]}] \right\} \\
 \Rightarrow &\sum_{m=1}^M \mathbb{I}[x^{[m]} \neq y^{[m]}] = \theta_e^{\text{ML}} \cdot M.
 \end{aligned} \tag{11}$$

To get from the second-to-last line to the last line, we used the fact that $x^{[m]}$ will either be the same as $y^{[m]}$ or different than $y^{[m]}$; i.e., $\mathbb{I}[x^{[m]} = y^{[m]}] + \mathbb{I}[x^{[m]} \neq y^{[m]}] = 1$.

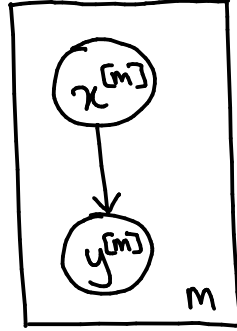
The left-hand side of the last line of Equation 11 essentially means that we have to count how many time $x^{[m]}$ and $y^{[m]}$ are different; i.e., $\sum_{m=1}^M \mathbb{I}[x^{[m]} \neq y^{[m]}] = \#(x^{[m]} \neq y^{[m]})$. After rearranging things, we can write the maximum-likelihood estimate as

$$\theta_e^{\text{ML}} = \frac{\#(x^{[m]} \neq y^{[m]})}{M}. \tag{12}$$

This will hopefully make intuitive sense to you: The maximum-likelihood estimate for the error probability is the proportion of the datapoints where the measurement was incorrect.

4 Example with Gaussian RVs

Let's work through another example, using the same plate model



but now using continuous random variables. The one factor is described by the marginal distribution

$$p\left(x^{[m]}\right) = \mathcal{N}\left(x^{[m]}; \mu_x, \sigma_x^2\right), \quad (13)$$

where the mean, μ_x , and the variance, σ_x^2 , are known parameters. The other factor, $p\left(y^{[m]}|x^{[m]}, \theta_{y|x}\right)$, models a noisy measurement of $x^{[m]}$ with an unknown offset, $\theta_{y|x}$, and is described by the affine transformation

$$y^{[m]} = x^{[m]} + \theta_{y|x} + v^{[m]}, \quad v^{[m]} \sim \mathcal{N}\left(v^{[m]}; 0, \sigma_v^2\right). \quad (14)$$

Here, the offset $\theta_{y|x}$ is the parameter to be estimated, and the other parameters are known. The parameters for this model to be estimated are therefore given by $\Theta = (\theta_{y|x})$. In preparation for the calculation of the maximum-likelihood estimate, we write the first factor of the model in terms of the exponential function:

$$p\left(x^{[m]}\right) = \frac{1}{\sqrt{2\pi\sigma_x^2}} \exp\left(-\frac{1}{2\sigma_x^2} \left(x^{[m]}\right)^2\right) = \exp\left(-\frac{1}{2\sigma_x^2} \left(x^{[m]}\right)^2 - \log \sqrt{2\pi\sigma_x^2}\right). \quad (15)$$

We write the second factor also in terms of the exponential function:

$$\begin{aligned} p\left(y^{[m]}|x^{[m]}, \theta_{y|x}\right) &= \mathcal{N}\left(y^{[m]}; x^{[m]} + \theta_{y|x}, \sigma_v^2\right) \\ &= \frac{1}{\sqrt{2\pi\sigma_v^2}} \exp\left(-\frac{1}{2\sigma_v^2} \left(x^{[m]} - y^{[m]} - \theta_{y|x}\right)^2\right) \\ &= \exp\left(-\frac{1}{2\sigma_v^2} \left(x^{[m]} - y^{[m]} - \theta_{y|x}\right)^2 - \log \sqrt{2\pi\sigma_v^2}\right). \end{aligned} \quad (16)$$

Since this second example has the same Bayes network structure as the first example (except for the name of the unknown parameter), we can use the decomposition developed in Equation 9, or

$$\log p(\mathcal{D}|\Theta) = \sum_{m=1}^M \left[\log p\left(y^{[m]}|x^{[m]}, \theta_{y|x}\right) + \log p\left(x^{[m]}\right) \right]. \quad (17)$$

We now combine Equations 15 to 17, which yields

$$\begin{aligned} \log p(\mathcal{D}|\Theta) &= \sum_{m=1}^M \left[\left(-\frac{1}{2\sigma_v^2} \left(x^{[m]} - y^{[m]} - \theta_{y|x}\right)^2 - \log \sqrt{2\pi\sigma_v^2} \right) \right. \\ &\quad \left. + \left(-\frac{1}{2\sigma_x^2} \left(x^{[m]}\right)^2 - \log \sqrt{2\pi\sigma_x^2} \right) \right]. \end{aligned} \quad (18)$$

At this point, we are ready to calculate the maximum-likelihood estimate of the parameter $\theta_{y|x}$, which we do in the same way as for the first example:

$$\begin{aligned}
 & \left. \frac{\partial}{\partial \theta_{y|x}} \log p(\mathcal{D}|\Theta) \right|_{\theta_{y|x} = \theta_{y|x}^{\text{ML}}} = 0 \\
 \Rightarrow & \sum_{m=1}^M \left[\left(\frac{1}{\sigma_v^2} (x^{[m]} - y^{[m]} - \theta_{y|x}^{\text{ML}}) - 0 \right) + 0 \right] = 0 \\
 \Rightarrow & \sum_{m=1}^M (x^{[m]} - y^{[m]}) = \sum_{m=1}^M \theta_{y|x}^{\text{ML}} \\
 \Rightarrow & \sum_{m=1}^M (x^{[m]} - y^{[m]}) = M \cdot \theta_{y|x}^{\text{ML}}.
 \end{aligned} \tag{19}$$

This means that the maximum-likelihood estimate is given by

$$\theta_{y|x}^{\text{ML}} = \frac{1}{M} \sum_{m=1}^M (x^{[m]} - y^{[m]}). \tag{20}$$

This result will hopefully also make sense to you: The offset parameter, $\theta_{y|x}$, is estimated to be the average of the difference between $x^{[m]}$ and $y^{[m]}$.